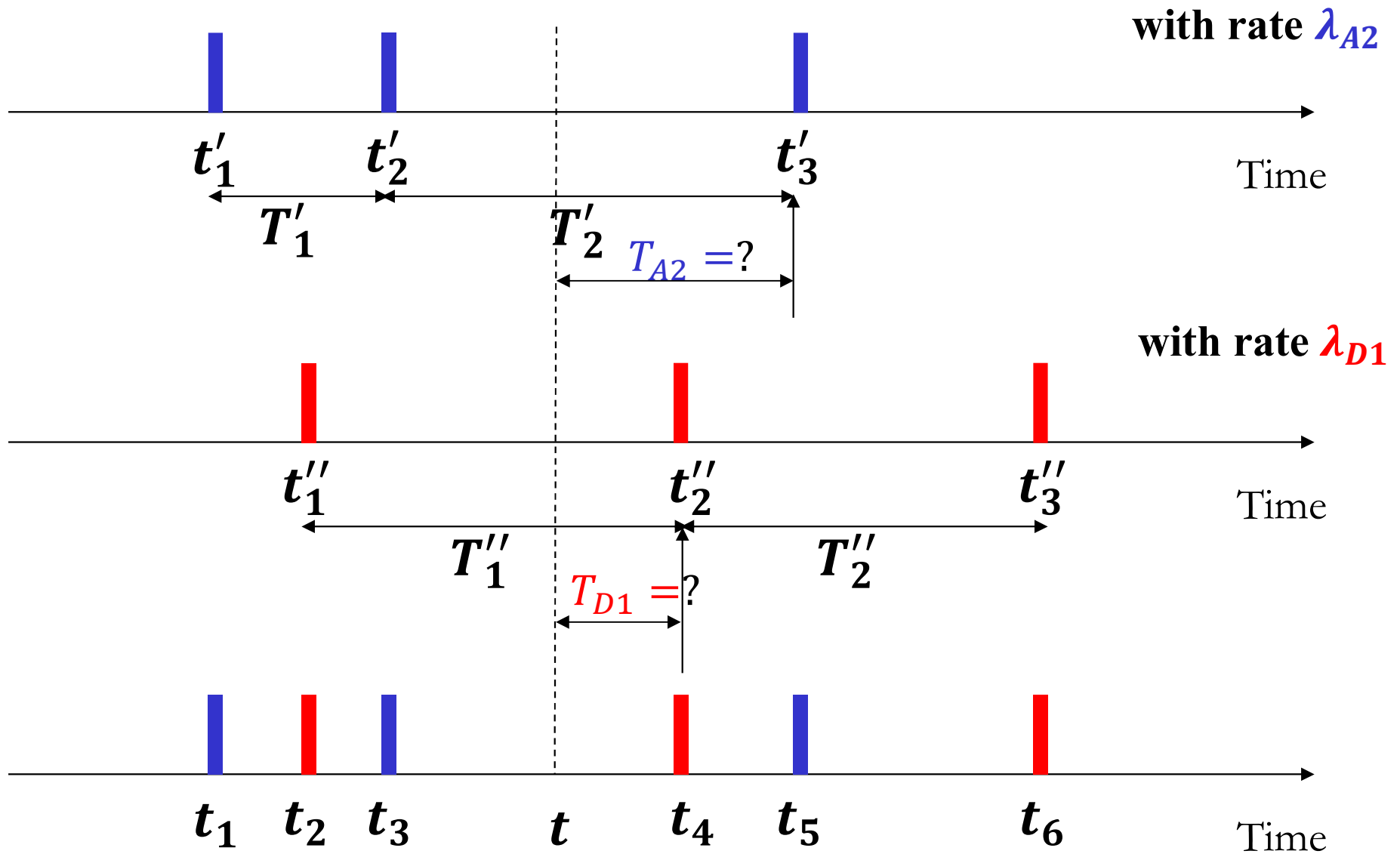


Tutorial 3

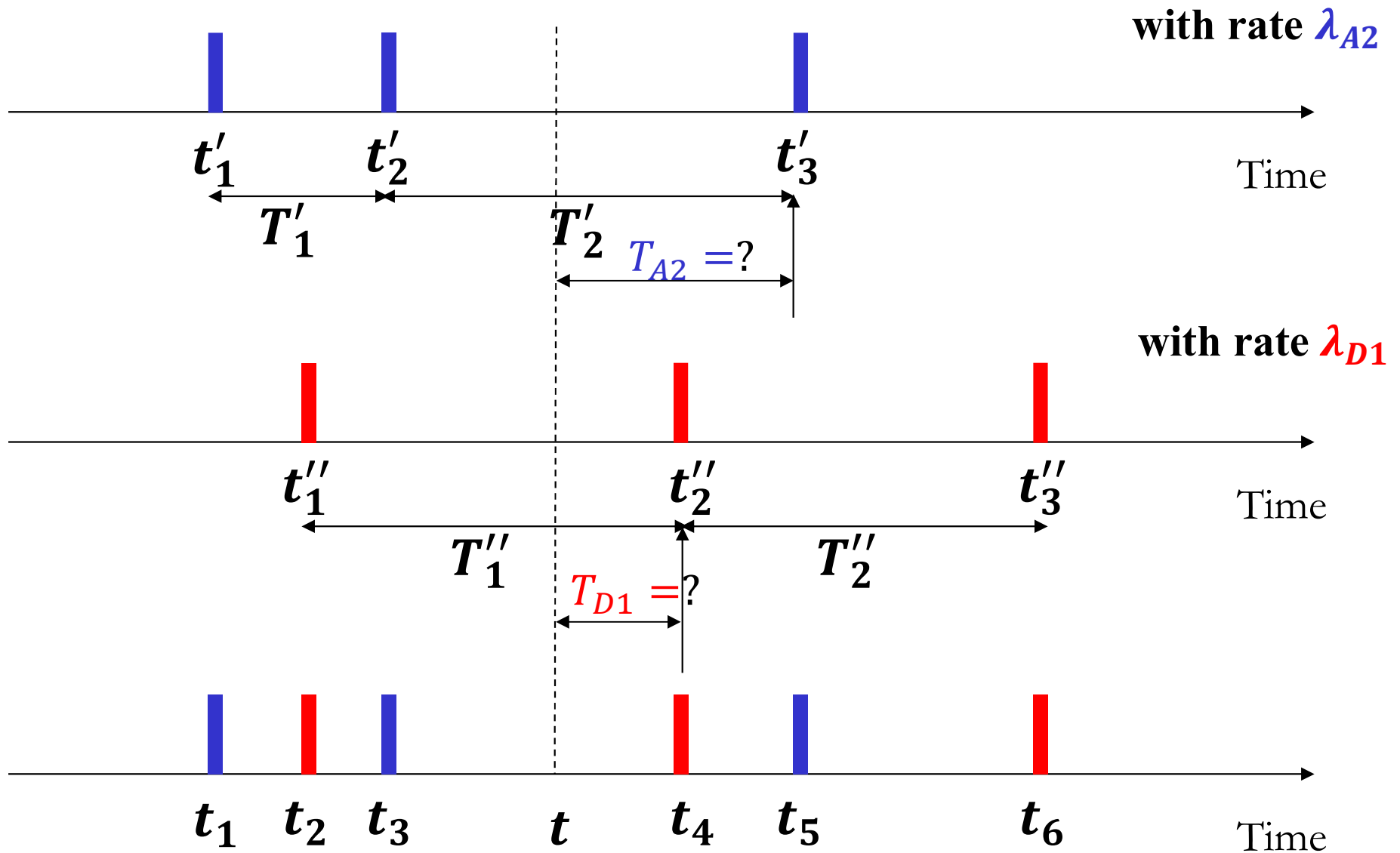
CS4226: Internet Architecture

Q1

- A2 and D1 buses arrive with rate λ_{A2} and λ_{D1} following Poisson processes. Suppose you will take the A2 or D1 bus that arrives the earliest. What is the probability that you will take an A2 bus?

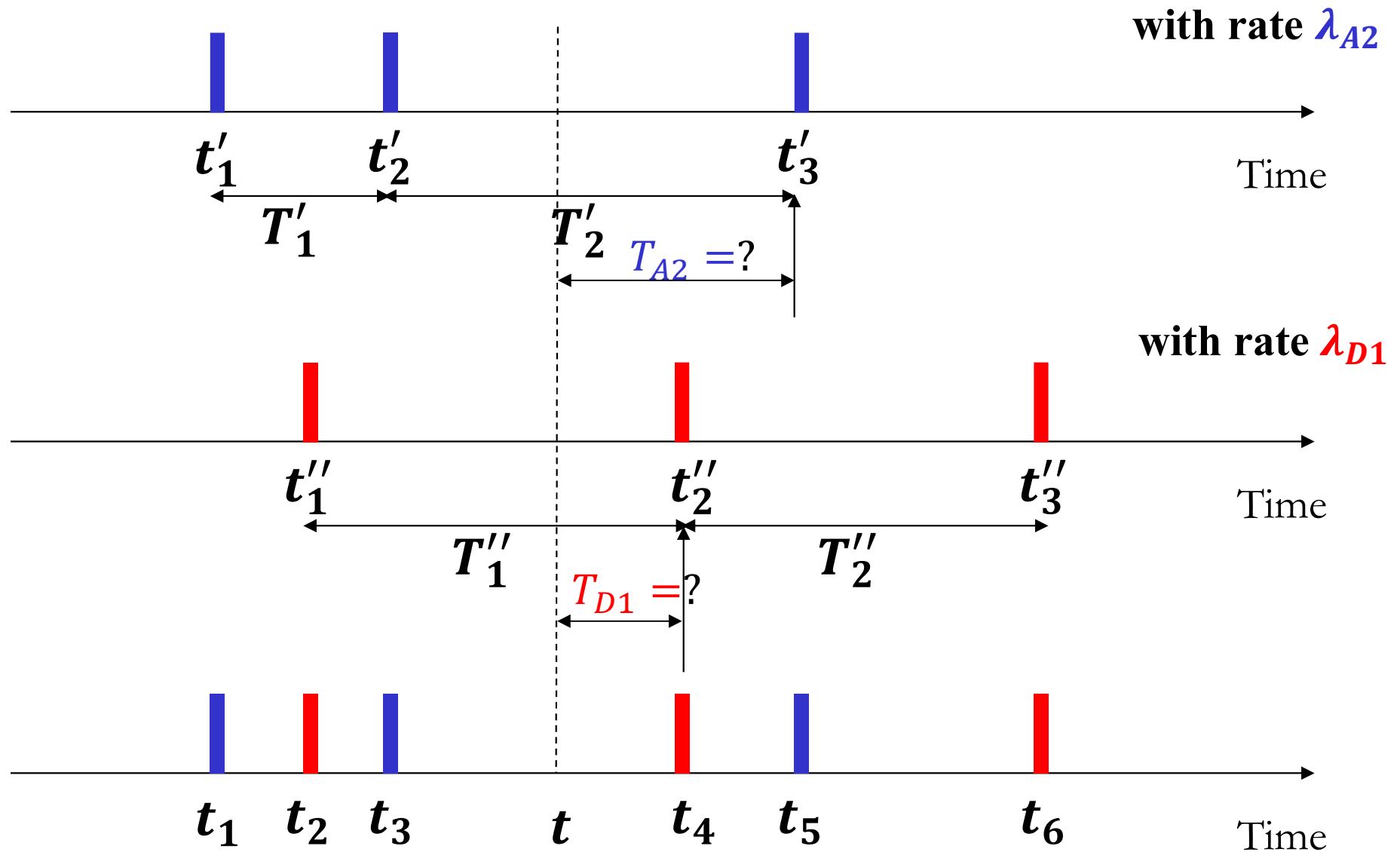


□ T_{A2} and T_{D1} are exp. r.v.s with rate λ_{A2} and λ_{D1}



□ T_{A2} and T_{D1} are exp. r.v.s with rate λ_{A2} and λ_{D1}

$$P\{T_{D1} > T_{A2}\}$$



□ T_{A2} and T_{D1} are exp. r.v.s with rate λ_{A2} and λ_{D1}

$P\{T_{D1} > T_{A2}\}$ How?

Law of Total Probability

□ Discrete distributions:

❖ For mutually exclusive events $E_1, E_2 \cdots E_n$

- $\sum_{i=1}^n P(E_i) = 1$

❖ Then:

- $P(E) = \sum_{i=1}^n P(E|E_i)P(E_i)$

□ Continuous distributions:

❖ For a random variable $Y \in [0, \infty]$.

- $\int_0^{\infty} f_Y(y) dy = 1$

❖ Then:

- $P\{X\} = \int_0^{\infty} P\{X|Y = y\} f_Y(y) dy$

Exponential Distribution

- A continuous r.v. T follows/has an **exponential distribution** with parameter $\lambda > 0$, if, for $x \geq 0$
 - ❖ $F(x) = P\{T \leq x\} = 1 - e^{-\lambda x}$ or $\bar{F}(x) = P\{T > x\} = e^{-\lambda x}$
 - ❖ $f(x) = \frac{dF(x)}{dx} = \lambda e^{-\lambda x}$
- $E[T] = \int_{-\infty}^{\infty} x f(x) dx = \frac{1}{\lambda}$
- What is the unit and physical meaning of λ ?
- ❖ Why is exponential a good distribution?
 - ❖ Memoryless property: $P\{T > s + t \mid T > s\} = P\{T > t\}$

Q1 (a rigorous proof)

- Let X and Y be two independent exponential random variables with parameters λ_x and λ_y . What is the probability $P\{X > Y\}$?

$$\begin{aligned} P\{X > Y\} &= \int_0^{\infty} P\{X > Y | Y = y\} f_Y(y) dy \\ &= \int_0^{\infty} P\{X > y\} (\lambda_y e^{-\lambda_y y}) dy = \int_0^{\infty} e^{-\lambda_x y} (\lambda_y e^{-\lambda_y y}) dy \\ &= \lambda_y \int_0^{\infty} e^{-(\lambda_x + \lambda_y)y} dy \\ &= \lambda_y \left(-\frac{1}{\lambda_x + \lambda_y} e^{-(\lambda_x + \lambda_y)y} \Big|_0^{\infty} \right) = \frac{\lambda_y}{\lambda_x + \lambda_y} \end{aligned}$$

Q1 (yet another way)

- Let X and Y be two independent exponential random variables with parameters λ_x and λ_y . What is the probability $P\{X > Y\}$?

$$\begin{aligned} P\{X > Y\} &= \int_0^{\infty} P\{X > Y | X = x\} f_X(x) dx \\ &= \int_0^{\infty} P\{Y < x\} (\lambda_x e^{-\lambda_x x}) dx = \int_0^{\infty} (1 - e^{-\lambda_y x}) (\lambda_x e^{-\lambda_x x}) dx \\ &= \lambda_x \int_0^{\infty} e^{-\lambda_x x} dx - \lambda_x \int_0^{\infty} e^{-(\lambda_x + \lambda_y)x} dx \\ &= \lambda_x \left(1 + \frac{1}{\lambda_x + \lambda_y} e^{-(\lambda_x + \lambda_y)x} \Big|_0^{\infty} \right) \\ &= 1 - \frac{\lambda_x}{\lambda_x + \lambda_y} = \frac{\lambda_y}{\lambda_x + \lambda_y} \end{aligned}$$

Q2

- Two Poisson packet flows with arrival rate $\lambda_1 = 5$ and $\lambda_2 = 3$ pkts/sec., going through a link with capacity of $\mu = 10$ pkts/sec. , modelled as $M/M/1$ system.
 - ❖ What is the average time for the packets to go through this link?
 - ❖ If we separate the link into μ_1 and μ_2 such that $\mu_1 + \mu_2 = \mu$, and serve flows separately. How do you allocate the capacity such that packets in both flows will spend the same amount of time going through the link? What is their average time going through the link?

Now you know all about M/M/1

□ Look at the system, i.e., server + queue

- ❖ Average # of packets in the system

$$E[L] = \frac{\rho}{1 - \rho}$$

- ❖ Average sojourn time of packets

$$E[W] = \frac{1}{\mu - \lambda}$$

□ Focus on the queue

- ❖ Average # of packets in the queue $E[Q] = \frac{\rho^2}{1 - \rho}$

- ❖ Avg. queueing delay of packets $E[D] = E[W] - \frac{1}{\mu}$

Now you know all about M/M/1

□ Look at the system, i.e., server + queue

- ❖ Average # of packets in the system

$$E[L] = \frac{\rho}{1 - \rho}$$

- ❖ Average sojourn time of packets

$$E[W] = \frac{1}{\mu - \lambda}$$

□ Focus on the queue

- ❖ Average # of packets in the queue $E[Q] = \frac{\rho^2}{1 - \rho}$

- ❖ Avg. queueing delay of packets $E[D] = E[W] - \frac{1}{\mu}$

Q2

- Two Poisson packet flows with arrival rate $\lambda_1 = 5$ and $\lambda_2 = 3$ pkts/sec., going through a link with capacity of $\mu = 10$ pkts/sec. , modelled as $M/M/1$ system.
 - ❖ What is the average time for the packets to go through this link?

Q2

- Two Poisson packet flows with arrival rate $\lambda_1 = 5$ and $\lambda_2 = 3$ pkts/sec., going through a link with capacity of $\mu = 10$ pkts/sec. , modelled as $M/M/1$ system.
 - ❖ What is the average time for the packets to go through this link?

$$E[W] = \frac{1}{\mu - (\lambda_1 + \lambda_2)} = \frac{1}{10 - 8} = \frac{1}{2};$$

Q2

- Two Poisson packet flows with arrival rate $\lambda_1 = 5$ and $\lambda_2 = 3$ pkts/sec., going through a link with capacity of $\mu = 10$ pkts/sec. , modelled as M/M/1 system.
 - ❖ If we separate the link into μ_1 and μ_2 such that $\mu_1 + \mu_2 = \mu$, and serve flows separately. How do you allocate the capacity such that packets in both flows will spend the same amount of time going through the link? What is their average time going through the link?

Q2

- Two Poisson packet flows with arrival rate $\lambda_1 = 5$ and $\lambda_2 = 3$ pkts/sec., going through a link with capacity of $\mu = 10$ pkts/sec. , modelled as $M/M/1$ system.
 - ❖ If we separate the link into μ_1 and μ_2 such that $\mu_1 + \mu_2 = \mu$, and serve flows separately. How do you allocate the capacity such that packets in both flows will spend the same amount of time going through the link? What is their average time going through the link?

$$E[W_1] = \frac{1}{\mu_1 - \lambda_1} = E[W_2] = \frac{1}{\mu_2 - \lambda_2}$$

Q2

- Two Poisson packet flows with arrival rate $\lambda_1 = 5$ and $\lambda_2 = 3$ pkts/sec., going through a link with capacity of $\mu = 10$ pkts/sec. , modelled as $M/M/1$ system.

$$E[W_1] = \frac{1}{\mu_1 - \lambda_1} = E[W_2] = \frac{1}{\mu_2 - \lambda_2}$$

$$\Rightarrow \begin{cases} \mu_1 - \lambda_1 = \mu_1 - 5 = \mu_2 - \lambda_2 = \mu_2 - 3 \\ \mu_1 + \mu_2 = \mu = 10 \end{cases} \Rightarrow \begin{cases} \mu_1 = 6 \\ \mu_2 = 4 \end{cases}$$

Q2

- Two Poisson packet flows with arrival rate $\lambda_1 = 5$ and $\lambda_2 = 3$ pkts/sec., going through a link with capacity of $\mu = 10$ pkts/sec. , modelled as $M/M/1$ system.

$$E[W_1] = \frac{1}{\mu_1 - \lambda_1} = E[W_2] = \frac{1}{\mu_2 - \lambda_2}$$

$$\Rightarrow \begin{cases} \mu_1 - \lambda_1 = \mu_1 - 5 = \mu_2 - \lambda_2 = \mu_2 - 3 \\ \mu_1 + \mu_2 = \mu = 10 \end{cases} \Rightarrow \begin{cases} \mu_1 = 6 \\ \mu_2 = 4 \end{cases}$$

$$\Rightarrow E[W_1] = E[W_2] = 1$$

Q3

- Suppose capacities μ_1 and μ_2 can be merged as $\mu = \mu_1 + \mu_2$.
 - ❖ Show that under $M/M/1$, for any two flows that can be supported by μ_1 and μ_2 , when you merge both and serve them by the aggregated capacity μ , the mean sojourn time is smaller than the minimum of the original mean sojourn times of the two packet flows.
 - ❖ Can we derive the same result for queueing delay?

Q3

$$E[W] < \min\{E[W_1], E[W_2]\}$$

Q3

$$E[W] < \min\{E[W_1], E[W_2]\}$$

$$E[W] = \frac{1}{\mu - (\lambda_1 + \lambda_2)} < \min \left\{ \frac{1}{\mu_1 - \lambda_1}, \frac{1}{\mu_2 - \lambda_2} \right\}$$

Q3

$$E[W] < \min\{E[W_1], E[W_2]\}$$

$$E[W] = \frac{1}{\mu - (\lambda_1 + \lambda_2)} < \min \left\{ \frac{1}{\mu_1 - \lambda_1}, \frac{1}{\mu_2 - \lambda_2} \right\}$$

$$E[W] = \frac{1}{\mu_1 - \lambda_1 + \mu_2 - \lambda_2} < \min \left\{ \frac{1}{\mu_1 - \lambda_1}, \frac{1}{\mu_2 - \lambda_2} \right\}$$

$$\mu_1 - \lambda_1 > 0$$

$$\mu_2 - \lambda_2 > 0$$

$$\mu_1 - \lambda_1 + \mu_2 - \lambda_2 > \max\{\mu_1 - \lambda_1, \mu_2 - \lambda_2\}$$

Q3

- Suppose capacities μ_1 and μ_2 can be merged as $\mu = \mu_1 + \mu_2$
 - ❖ Can we derive the same for queueing delay?

Q3

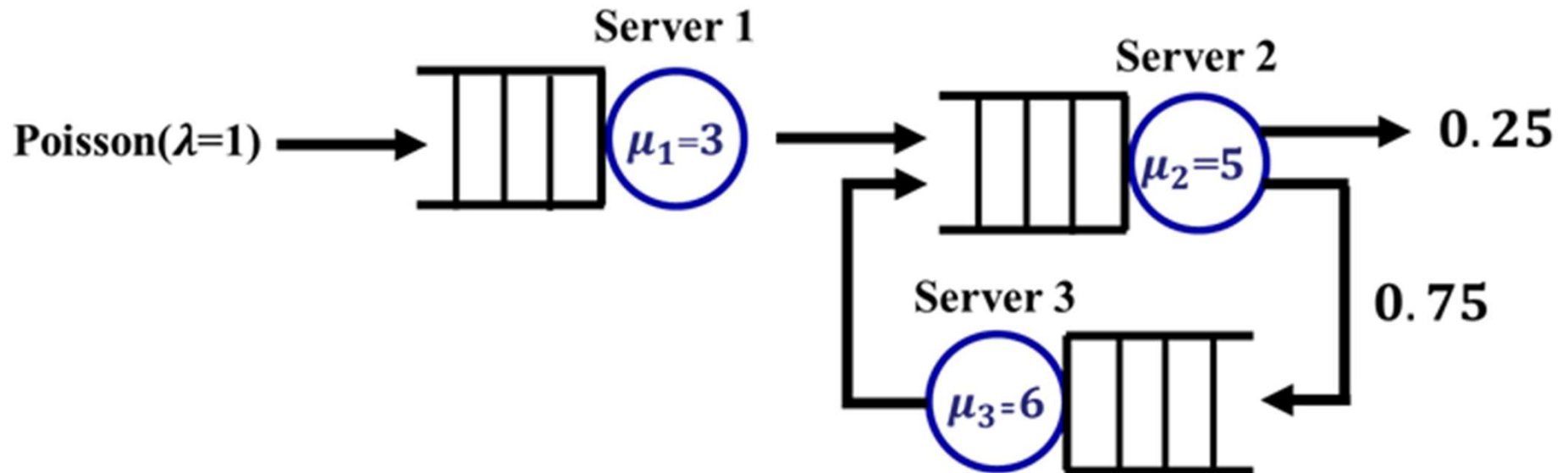
Consider $\lambda_1 \rightarrow 0$, $E[D] = E[W] - \frac{1}{\mu}$

$E[D] > \min\{E[D_1], E[D_2]\}$ *because:*

$$\frac{1}{\mu - (\lambda_1 + \lambda_2)} - \frac{1}{\mu} > \min \left\{ \frac{1}{\mu_1 - \lambda_1} - \frac{1}{\mu_1}, \frac{1}{\mu_2 - \lambda_2} - \frac{1}{\mu_2} \right\}$$

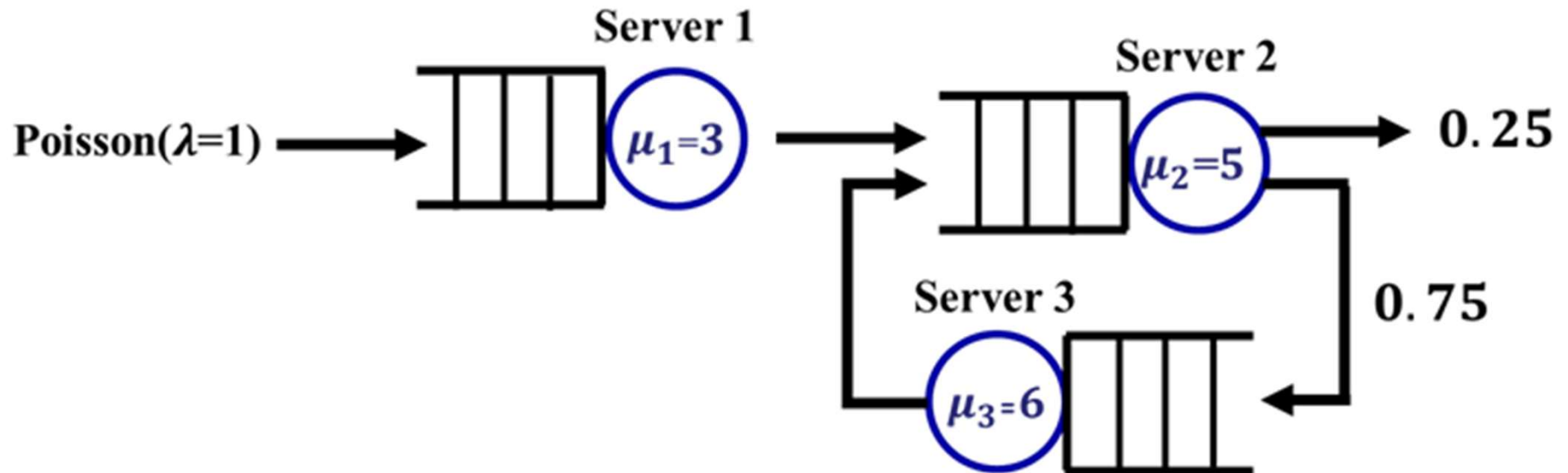
$$\frac{1}{\mu - \lambda_2} - \frac{1}{\mu} > \min \left\{ 0, \frac{1}{\mu_2 - \lambda_2} - \frac{1}{\mu_2} \right\}$$

Q4



- In the above Jackson network, what is the average sojourn time of the packets, $E[W]$?

Q4



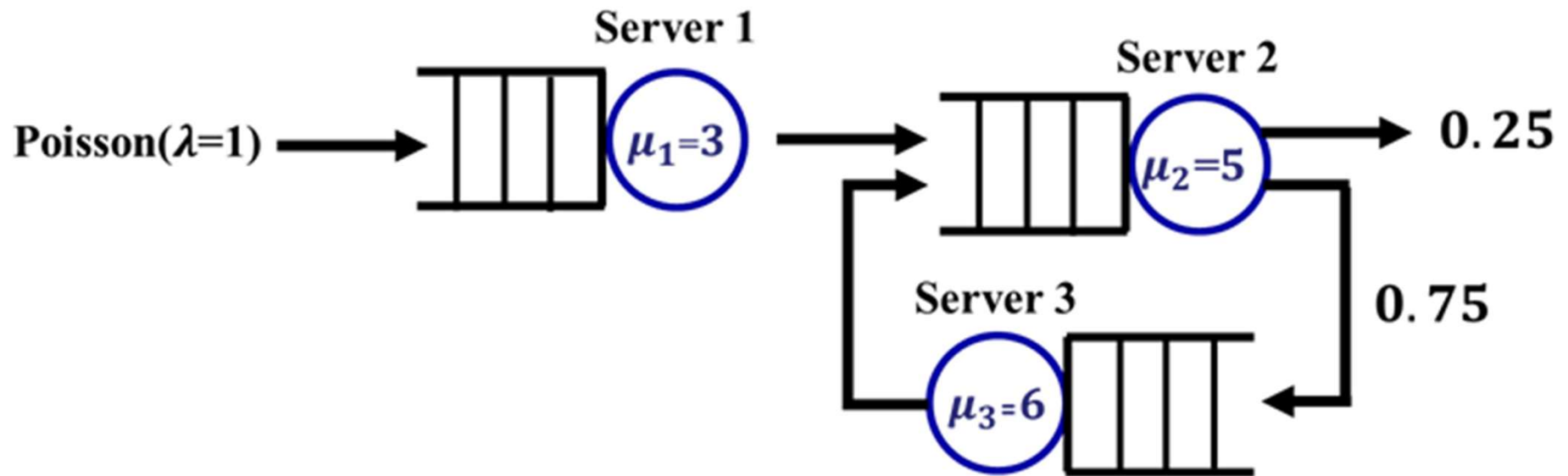
□ A two-step procedure

- ❖ 1. Determine the utilization ρ_i for each server i .
- ❖ 2. Apply the formula to get the average sojourn time

Q4

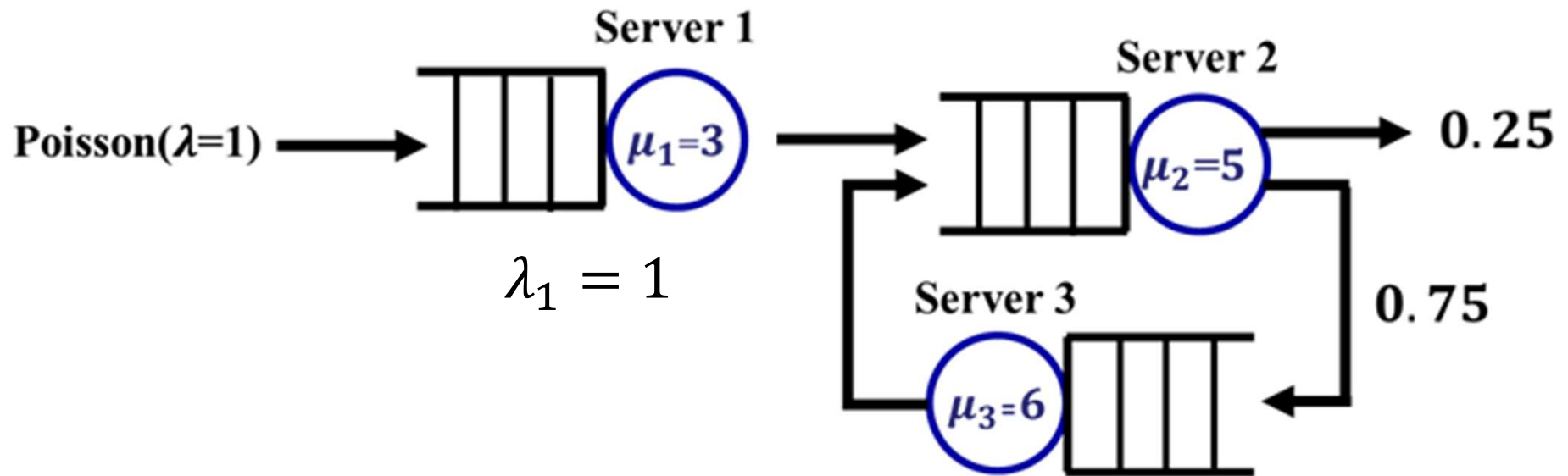
❖ Calculate effective arrival from external arrival and feedback arrivals from other subsystems.

$$\lambda_i = r_i + \sum_{j=1}^n \lambda_j P_{ji}$$



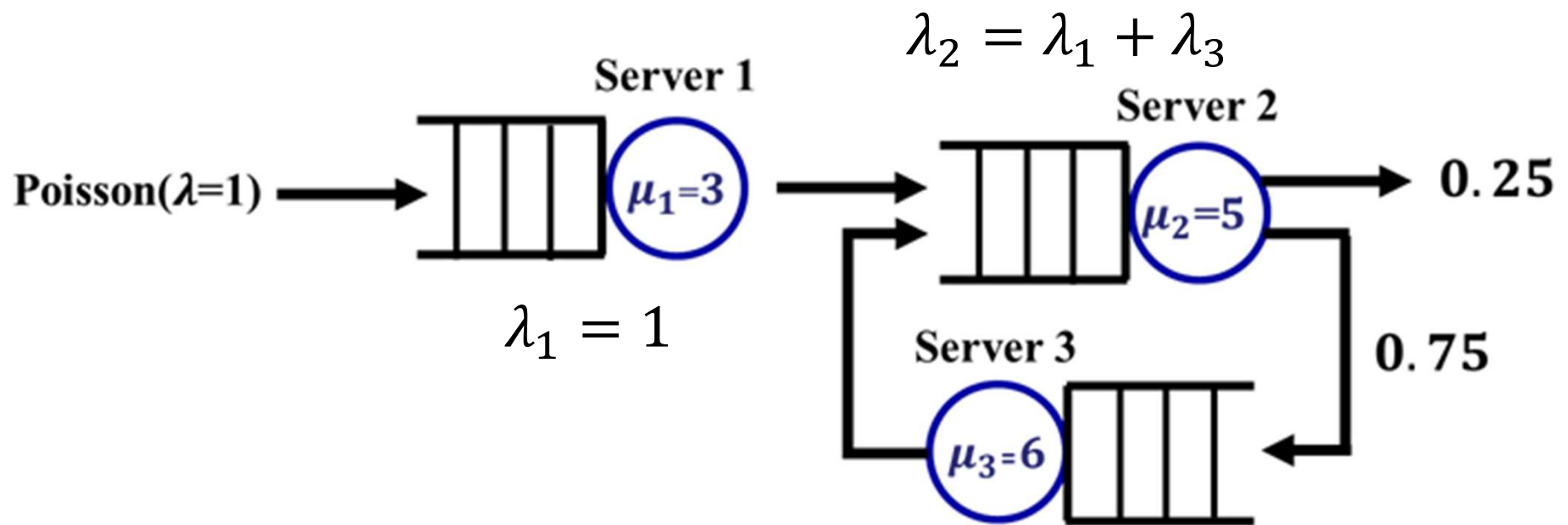
Q4

- ❖ Calculate effective arrival from external arrival and feedback arrivals from other subsystems.



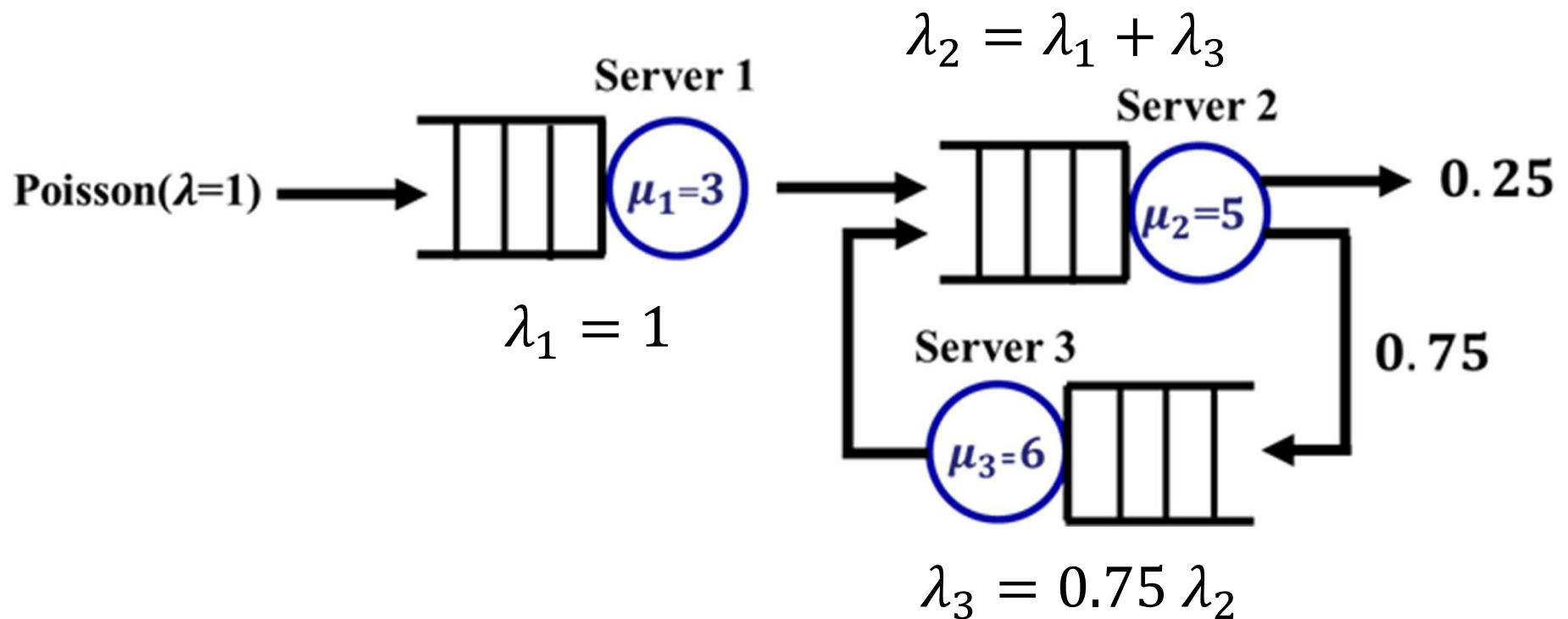
Q4

- ❖ Calculate effective arrival from external arrival and feedback arrivals from other subsystems.



Q4

- ❖ Calculate effective arrival from external arrival and feedback arrivals from other subsystems.



Q4

1. Determine the utilization ρ_i for each server i .

$$\begin{cases} \mu_1 = 3 \\ \mu_2 = 5 \\ \mu_3 = 6 \end{cases} \quad \begin{cases} \lambda_1 = ? \\ \lambda_2 = ? \\ \lambda_3 = ? \end{cases}$$

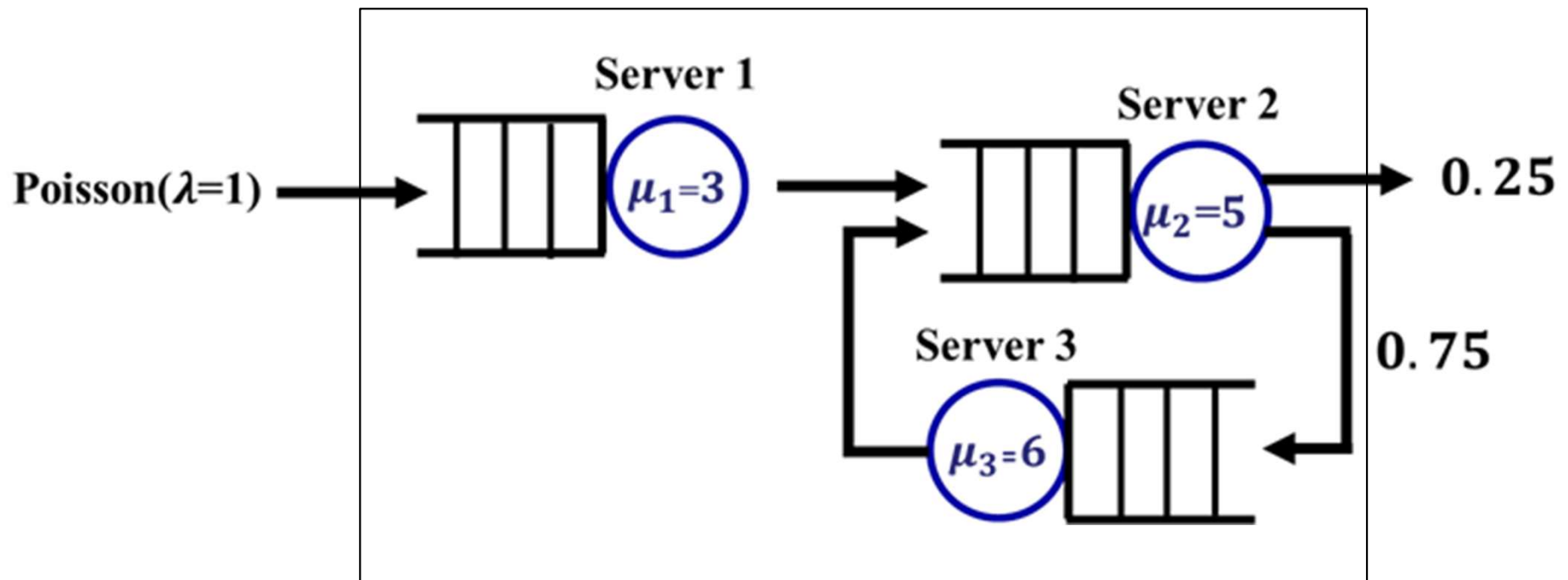
$$\begin{cases} \lambda_1 = 1 \\ \lambda_2 = \lambda_1 + \lambda_3 \\ \lambda_3 = 0.75 \lambda_2 \end{cases} \Rightarrow \begin{cases} \lambda_1 = 1 \\ \lambda_2 = 1 + 0.75 \lambda_2 \\ \lambda_3 = 0.75 \lambda_2 \end{cases} \Rightarrow \begin{cases} \lambda_1 = 1 \\ \lambda_2 = 4 \\ \lambda_3 = 3 \end{cases}$$

$$\Rightarrow \begin{cases} \rho_1 = 1/3 \\ \rho_2 = 4/5 \\ \rho_3 = 3/6 = 1/2 \end{cases}$$

Q4

□ Method 1:

- ❖ Calculate average number of packets in the whole system, and apply Little's Law.



Q4

□ Method 1:

- ❖ Calculate average number of packets in the whole system, and apply Little's Law.

$$E[L_i] = \frac{\rho_i}{1 - \rho_i}$$

$$E[L] = E[L_1] + E[L_2] + E[L_3]$$

$$\lambda = \lambda_1$$

$$E[L] = \lambda E[W]$$

Q4

□ Method 1:

- ❖ Calculate average number of packets in the whole system, and apply Little's Law.

$$E[L_1] = 1/2; E[L_2] = 4; E[L_3] = 1$$

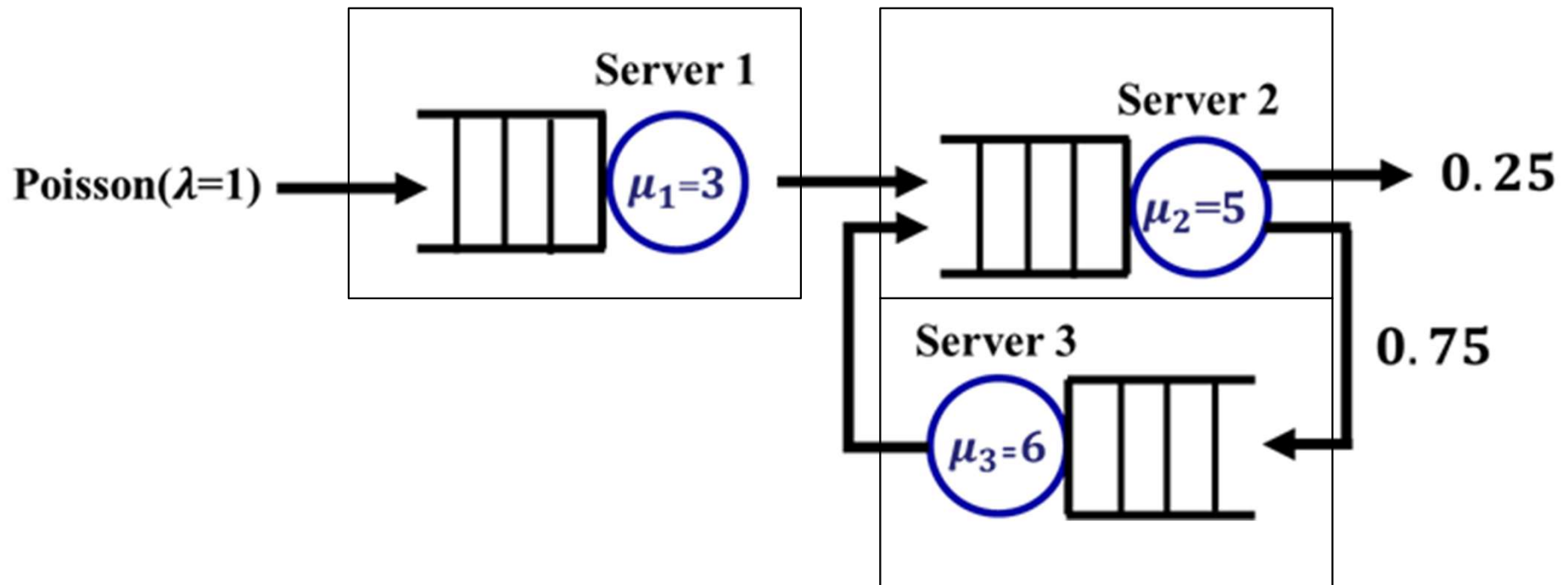
$$\lambda = \lambda_1 = 1$$

$$E[W] = \frac{E[L]}{\lambda} = 5.5$$

Q4

□ Method 2:

- ❖ Sum up average sojourn time of packets in each subsystem.



Q4

□ Method 2:

- ❖ Sum up average sojourn time of packets in each subsystem.

Let V_i be the r.v. to indicate that number of loops a packet spends in a subsystem i .

$$E[W_i] = \frac{1}{\mu_i - \lambda_i}$$

$$E[W] = E[V_1]E[W_1] + E[V_2]E[W_2] + E[V_3]E[W_3]$$

Q4

Let V_i be the r.v. to indicate that number of loops a packet spends in a subsystem i .

$$P(V_1 = 1) = 1 \rightarrow E(V_1) = 1$$

Q4

Let V_i be the r.v. to indicate that number of loops a packet spends in a subsystem i .

$$P(V_1 = 1) = 1 \rightarrow E(V_1) = 1$$

$$E(V_2) = E(V_3) + 1$$

Q4

Let V_i be the r.v. to indicate that number of loops a packet spends in a subsystem i .

$$P(V_1 = 1) = 1 \rightarrow E(V_1) = 1$$

$$E(V_2) = E(V_3) + 1$$

$$P(V_3 = i) = 0.75^i(1 - 0.75) \rightarrow$$

$$\begin{aligned} E[V_3] &= 0 * 0.25 + 1 * 0.75 * 0.25 + 2 * 0.75^2 * 0.25 + \dots \\ &= \frac{p}{1 - p} = \frac{0.75}{1 - 0.75} = 3 \end{aligned}$$

Q4

□ Method 2:

- ❖ Sum up average sojourn time of packets in each subsystem.

$$E[W_1] = \frac{1}{2}; E[W_2] = 1; E[W_3] = \frac{1}{3}$$

$$E[V_1] = 1; E[V_2] = 4; E[V_3] = 3$$

$$\begin{aligned} E[W] &= E[V_1]E[W_1] + E[V_2]E[W_2] + E[V_3]E[W_3] \\ &= 5.5 \end{aligned}$$

Thank You